STA572's GROUP ASSIGNMENT 2 (20%)

Univariate Time Series Modeling and Forecasting of Global Oil Production (1990–2023) Using Exponential Smoothing (ETS) and ARIMA-Based Approaches

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Abstract:

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1. Introduction

The global oil industry continues to be one of the most significant areas of the energy economy. It is simultaneously both a driver of growth and a reflection of geopolitical and environmental changes. Data on oil production, particularly with a global dimension, is a vital indicator of energy security, economic stability, and transition readiness (Macovei, 2023; Shahbaz, 2024). Whereas traditional econometric studies typically focused on the multivariate factors underlying oil production, particularly demand shocks, crude oil price shocks, changes to energy policy, or global crises, there has been less emphasis on univariate time series models which can more easily utilize historical data trends for long-term use in forecasting and planning.

When the predictors external to the process are unreliable, unavailable, or non-stationary, univariate time series analysis may be more appropriate for predictive analysis. Given the presence of long historical records for global oil production (1990–2023), statistical modeling can lead to useful forecasts through the use of exponential smoothing (ETS) and Box-Jenkins ARIMA models (Cabral Marques & Barros Filho, 2024). As seen with ETS and ARIMA, both have become quite standard for energy and oil forecasting, with exponential smoothing known to be robust, even with relatively noisy data, and ARIMA known for flexibility for autoregressive and moving average behavior (Hyndman & Athanasopoulos, 2021).

Recent studies support this as Marques et al. (2024) found that both exponential smoothing and ARIMA could effectively model oil prices, with ARIMA out-performing during the volatile periods. Hybrid studies also sought to improve the forecasting from traditional time series models by either neural networks or grey models; however, these models tend to have lower accuracy but require more granular data, and they must be more complicated to tune (Yin et al., 2025; Bai et al., 2024). In terms of structured forecast models that can be interpreted, conventional ETS and ARIMA techniques continue to be industry benchmarks (Liu et al., 2022; Xie et al., 2024).

This study will generate a robust and interpretable forecasting model of global oil production, using univariate time series techniques. The focus of Part A of the study will incorporate various ETS models, like the Naïve, Average, Holt's Linear trend, and Holt-Winters models, and identify the model with the best performance using error statistics like RMSE, MAE, and MAPE. Part B will apply the Box-Jenkins approach evaluating an AR, MA, ARMA, and ARIMA model, and test each model for stationarity, independence of residuals, and normality. The two approaches provide a comparative assessment of the best-performing univariate model to forecast the production of oil until 2035.

The relevance of this research is highlighted by energy transition debates, climate mitigation targets, and rising uncertainty in global oil demand due to economic or external shocks such as COVID-19 (Xu et al., 2023; Sampene et al., 2024). Oil production based forecasting underpins energy economic forecasting but more broadly it informs investment and infrastructure, and globally, sustainability policy. Furthermore, using endogenous historical data mitigates the challenges of multicollinearity and data gaps from use of multivariate representation in macroeconomic modelling (Tillaguango et al., 2024; Rehman & Ahmad, 2024).

1.1 Research Objectives

This study aims to model and forecast the global oil production trend covering the period 1990-2023, through univariate time series analysis. The univariate techniques employed were exponential smoothing (ETS) and Box-Jenkins ARIMA-based models. Whereas multivariate regression methodologies are based on multiple variables, univariate only relies on historical oil production to establish all the dynamics and temporal dependencies. The emphasis of the research was to compare the performance of models, and to identify the best forecasting technique for projecting into the future. The research objectives of this study are as follows:

1. To study the trend and trajectory of global oil production over time using historical time series data.
2. To determine the most appropriate univariate time series models for oil production forecasting based on model fit and diagnostic evaluation.
3. To develop a reliable and interpretable forecasting model for global oil production using exponential smoothing and ARIMA-based methods.

2. Description of Data

This study utilizes yearly global oil production data, reported in terawatt-hours (TWh), obtained from Our World in Data and based on the Energy Institute's Statistical Review of World Energy (2024). It includes annual observations from 1990 to 2023, comprising a total of 34 yearly observations. The analysis is performed univariately, as oil production is the only time series of interest. This dataset highlights the long-run oil production trends driven by global economic cycles, energy transitions and exogenous shocks, such as pandemics and geopolitical disruption. As noted in Figure 2.1, global oil production has an overall upward trend, with variation across decades. The study utilizes univariate forecasting techniques based on time series analysis ETS and ARIMA-based models to identify underlying production patterns and produce valid future forecasts, without accounting for exogenous explanatory variables.

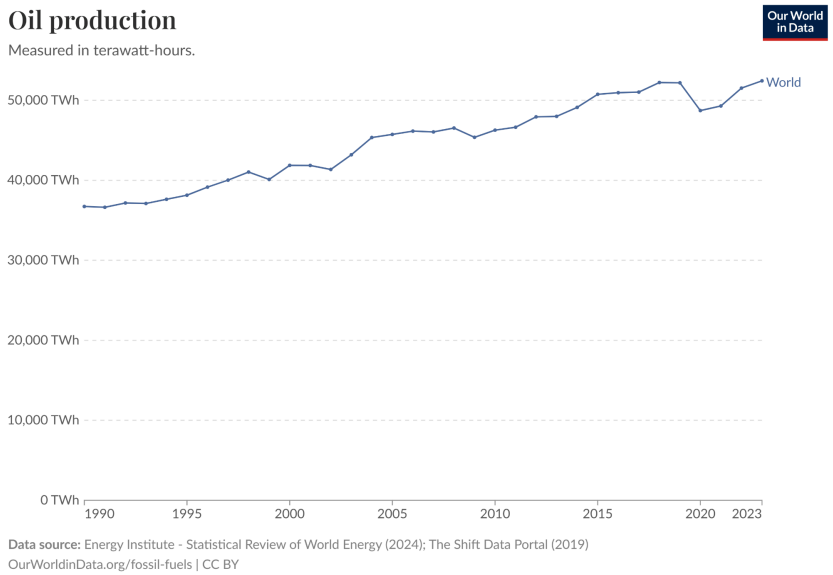


Figure 2.1 Oil production

In preparation for univariate time series analysis, the dataset was imported using the read.csv() function. The filename, Assignment 2 dataset (Oil Production 1990-2023).csv, included annual oil production figures for various entities. A screenshot of the output produced in the R console is captured in Figure 2.2 for documentation purposes.

```
> # Read CSV file
> time_series = read.csv("Assignment 2 dataset (Oil Production 1990-2023).csv")
> summary(time_series)
    Entity              Code                Year      Oil.production..TWh.
 Length:34          Length:34          Min.   :1990   Min.   :36628
 Class :character   Class :character   1st Qu.:1998   1st Qu.:40337
 Mode  :character   Mode  :character   Median :2006   Median :45889
                                       Mean   :2006   Mean   :44827
                                       3rd Qu.:2015   3rd Qu.:49010
                                       Max.   :2023   Max.   :52432
```

Figure 2.2 Importing the oil production time series into R

The summary statistics in Figure 2.3 show the minimum (36,628), maximum (52,432), mean (44,827), median (45,889), standard deviation (5,165.57) and skewness (–0.15). We have 34 observations and no missing values. The results are tabulated in Table 2.1 below. The first and last six rows of the dataset are also available through the head() and tail() functions. This initial inspection indicates a roughly symmetric distribution without any missing values, creating a data set that we can confidently use for time series modelling.

```
> #step 2
> std_dev = sd(time_series$Oil.production..TWh.)
> std_dev
[1] 5165.57
> skew = skewness(time_series$Oil.production..TWh.)
> skew
[1] -0.1553661
> missing = sum(is.na(time_series$Oil.production..TWh.))
> missing
[1] 0
> summary(time_series)
    Entity              Code                Year      Oil.production..TWh.
 Length:34          Length:34          Min.   :1990   Min.   :36628
 Class :character   Class :character   1st Qu.:1998   1st Qu.:40337
 Mode  :character   Mode  :character   Median :2006   Median :45889
                                       Mean   :2006   Mean   :44827
                                       3rd Qu.:2015   3rd Qu.:49010
                                       Max.   :2023   Max.   :52432
> head(time_series)
  Entity     Code Year Oil.production..TWh.
1  World OWID_WRL 1990             36726.78
2  World OWID_WRL 1991             36627.71
3  World OWID_WRL 1992             37158.00
4  World OWID_WRL 1993             37105.16
5  World OWID_WRL 1994             37619.20
6  World OWID_WRL 1995             38135.52
> tail(time_series)
   Entity     Code Year Oil.production..TWh.
29  World OWID_WRL 2018             52221.39
30  World OWID_WRL 2019             52188.84
31  World OWID_WRL 2020             48708.86
32  World OWID_WRL 2021             49286.68
33  World OWID_WRL 2022             51519.52
34  World OWID_WRL 2023             52432.23
```

Figure 2.3 Summary statistics, standard deviation, skewness, and missing values of time_series

Table 2.1 Summary of Descriptive Statistics for Global Oil Production Time Series

Description (Abbr.)	Length	Min	Med	Mean	Max	Sd	Sk	NA
Global Oil Production (TWh)	34	36628.00	45889.00	44827.00	52432.00	5165.57	-0.1534	0

To evaluate the distribution shape of oil production values, we created both a histogram and a boxplot (figure 2.4). The histogram indicates a slightly left-skewed distribution, which is something that the skewness value (-0.15) supports. The box plot suggests a fairly symmetrical spread of values around the median. Similarly, the boxplot stats() function indicated there were no outliers, as displayed in figure 2.5.

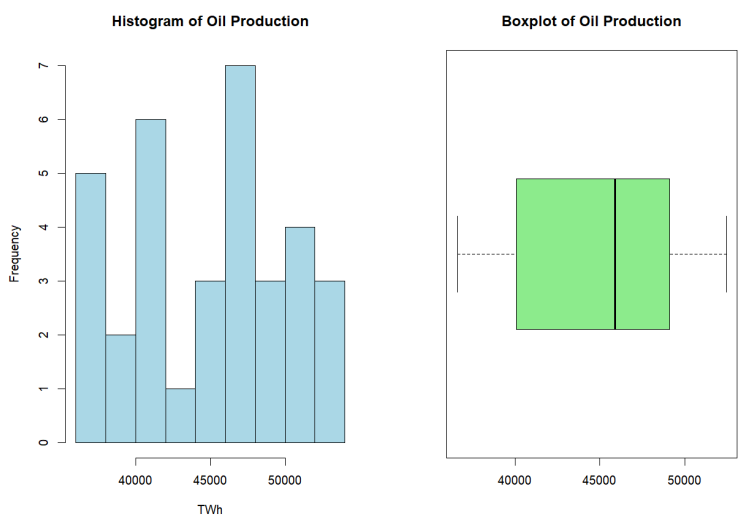


Figure 2.4 Histogram & Boxplot of the time series data

```
> outliers = boxplot.stats(time_series$Oil.production..TWh.)$out
> outliers
numeric(0)
```

Figure 2.5 Potential Outliers

The time series plot of global oil production from 1990 to 2023 is provided in Figure 2.6. A clear upward trend is present, with occasional short-term fluctuations around 2008 and 2020. Thus, a trend component is demonstrated. The length() function in Figure 2.7 was used to calculate the length(DV)), indicating the time series includes 34 annual observations in the periods stated.

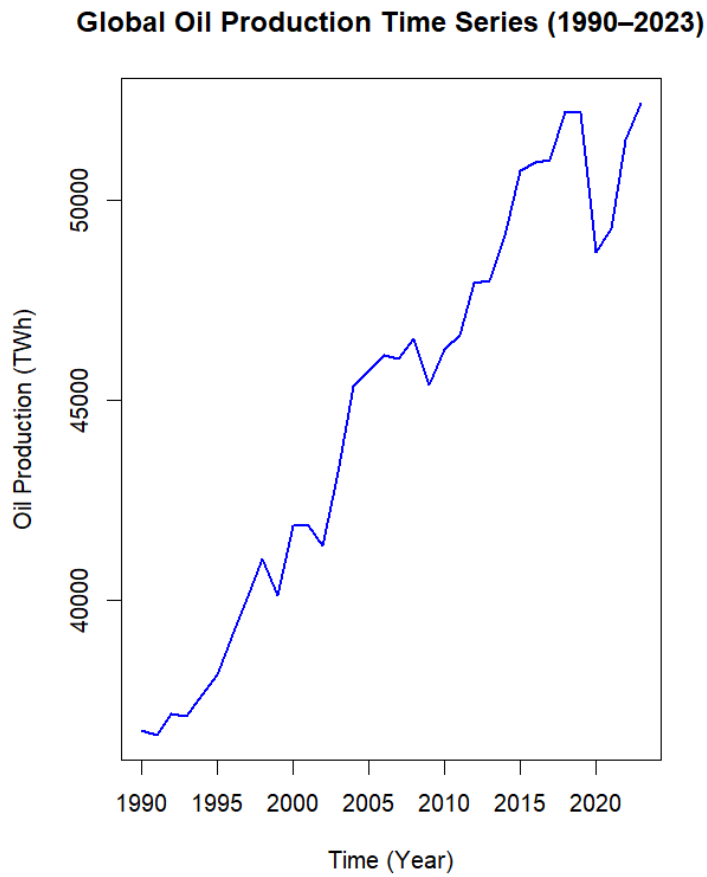


Figure 2.6 Time series plot of global oil production

```
> length(DV)
[1] 34
```

Figure 2.7 Output of length(DV)

Figure 2.8 shows the autocorrelation function (ACF) plot, which shows the autocorrelation decays slowly, again confirming a trend and no white noise. Finally, the ggseasonplot() function returned an error, indicating no seasonality is present in the dataset, which is confirmed in Figure 2.9.

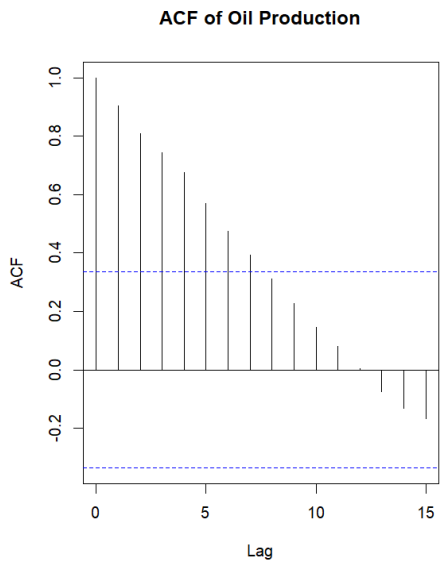


Figure 2.8 Autocorrelation function (ACF) plot of the time series data

```
> oil_ts = ts(time_series$Oil.production..TWh., start = 1990, frequency = 1)
> ggseasonplot(oil_ts)
Error in ggseasonplot(oil_ts) : Data are not seasonal
```

Figure 2.9 Error from ggseasonplot()

3. Methodology

PART A (ETS)

Model 1: Naïve with Drift

The Naïve with Drift model is a straightforward and effective forecasting model. In this model, the basic naïve model is extended to include a linear drift term. The model assumes that the difference between consecutive observations is constant over time and therefore projects a linearly increasing or decreasing trend into the future. The model is written as:

$$\hat{y}_{T+h|T} = y_T + \frac{h}{T-1}\sum_{t=2}^{T}(y_t - y_{t-1})$$

Where $\hat{y}_{T+h|T}$ is the forecasted value for $T + h$, and y_T is the last observed value at time T (Lazim, 2021). The model is practical because it does not require much processing power and involves little parameter tuning. It is often treated as a benchmark model to compare against more complex approaches. Hasri et al. (2021) applied the Naïve model for evaluation of daily COVID-19 cases in Malaysia, where they demonstrated that the Naïve model was very powerful for short-term forecasting without much complexity.

Model 2: Holt's Linear Trend Method

Holt's Linear Trend Model, or double exponential smoothing, was introduced by Charles Holt in 1957 as an expansion of simple exponential smoothing, allowing for forecasting of data with a linear trend. The model has two smoothing equations, one for level ℓ_t and one for trend b_t :

$$\hat{y}_{t+h|t} = \boldsymbol{\ell_t} + \boldsymbol{hb_t}$$

$$\boldsymbol{\ell_t} = \boldsymbol{ay_t} + (1-\alpha)(\boldsymbol{\ell_{t-1}} + \boldsymbol{b_{t-1}})$$
$$\boldsymbol{b_t} = \beta(\boldsymbol{\ell_t} - \boldsymbol{\ell_{t-1}}) + (1-\beta)\boldsymbol{b_{t-1}}$$

where α and β are smoothing constants between 0 and 1.

The model is known for predicting slow changes in trend behaviour in a series, and is particularly well suited for economic, energy, and transportation data. Sibuea et al. (2022) compared Holt's Linear Trend Model and Linear Regression for the forecasting of oil and gas exports in Indonesia, finding Holt's model had lower forecasting error and is more effective for short term forecasting.

Model 3: Average Change Method

The Average Change Method forecasts future values by calculating the average change in history in the time series. This method assumes that the average change in the historical data will be repeated in the future. The generic formula for the forecast is as follows :

$$\hat{y}_{t+h} = y_1 + h \cdot \bar{\Delta}, \quad \text{where } \bar{\Delta} = \frac{1}{n-1} \sum_{t=2}^{n} (y_t - y_{t-1})$$

This approach is computationally simple, and is appropriate for datasets that have significant long-term monotonic trends. Banditvilai and Araveeporn (2024) used similar simple models, including average extrapolation methods, to forecast crude palm oil production in Thailand, and found these models to be useful when seasonality was not a factor. Martinez-Prieto et al. (2024) provided evidence supporting the use of average-trend forecasting when undertaking long-horizon oil demand modeling for OECD and non-OECD regions. Fattah et al. (2018) demonstrated that mean-difference methods provide a reasonable and admissible baseline to compare with more complex models used in energy consumption forecasting.

PART B
Model 4: ARIMA

The Autoregressive Integrated Moving Average (ARIMA) model is a well-known statistical technique for analyzing and forecasting univariate time series. It is especially useful for non-stationary data with trends and autocorrelation (Hyndman & Athanasopoulos, 2021). The notation for the ARIMA model is ARIMA(p,d,q), where p is the number of autoregressive (AR) terms, d is the number of differences taken to

achieve stationarity, and q is the number of moving average (MA) terms. The general representation of an ARIMA(p,d,q) model is as follows:

$$Y_t = c + \sum_{i=1}^{p} \phi_i Y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \epsilon_t$$

Where Y_t t is the observed value at time *t*, c is a constant term, ϕ\phi_i are autoregressive coefficients, θ\theta_j are moving average coefficients, and ε_t represents white noise errors (Hyndman & Athanasopoulos, 2021).

ARIMA models have been shown to perform well in a number of applications in practice. One study that employed ARIMA successfully is Wahyudi & Sari (2024), which applied ARIMA to forecast stock price changes in the Indonesian capital market. These methods are also utilized in the energy sector when forecasting and simulating investments in the renewable energy industry. The general purpose of ARIMA's applicability to sustainability planning and energy forecasting is substantiated by Mulyono and Rahman (2023) as they applied the model to forecast solar stock.

For this research, yearly data from Our World in Data was used to forecast global oil production from 1990 to 2023 in the ARIMA model. The Box-Jenkins (1976) methodology involves four processes in the modeling process including model identification, parameter estimation, diagnostic checking, and forecastings. The first process is to determine whether the data is stationary. If the series displays trends or non-constant variance, differentiating is applied until the data is made stationary. After the data is appropriately differentiated, the information provided from the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots are used to identify reasonable values for the AR and MA components; these plots provide valuable information to correctly select the lag orders based on the structure of the time series.(Marques & Barros Filho, 2024)

Following the identification process, the parameters of the ARIMA models were estimated via maximum likelihood methods. The purpose is to find the AR and MA coefficient values that best represent the behavior of the differenced series. Next, residual diagnostics are conducted to examine the fit of the models. The residuals ideally should represent white noise, specifically, stationary with constant variance, no

autocorrelation, and approximately normally distributed. This is usually accomplished through visual inspection of the residuals plots and through statistical tests such as the Ljung-Box test.

The model can be used for forecasting after it passes the diagnostic tests. The ability to represent multiple time series patterns by ARIMA models, including non-seasonal time series patterns, makes them especially useful. ARIMA models are widely used in financial, economic and industrial contexts for short to medium-term forecasting (Marques & Barros Filho, 2024; Liu et al., 2022).

Within this study, ARIMA was used as a benchmark in the forecasting of global oil production. ARIMA is a useful method in this domain that provides a way to explore the temporal characteristics of oil production data and forecast based on previous observations that is systematic and statistical.

4. Analysis and Results

4.1 Analysis on P-value Assumptions

Table 4.1 p-Value for the assumptions of normality, independence, homoskedasticity, and zero autocorrelation for the residuals of the fitted models

Residuals	**p-value**			
	ad.test	box.test	dw.test	bp.test
PART A (ETS)				
Naïve with Drift	0.041	0.9321	4.2e-01	0.1168
Holt's Linear Trend Method	0.024	0.9087	3.5e-01	0.0983
Average Change Method	0.275	0.0002	2.7e-05	0.0093
PART B				
AR(1)	0.0192	0.5843	0.3709	0.1642
ARMA(1,0,1)	0.0416	0.6438	0.7242	0.1885
ARIMA(1,1,1)	0.0078	0.5382	0.8216	0.1216
Auto ARIMA (0,1,0)	0.0025	0.3844	0.5284	0.0996

For Part A, The Anderson-Darling (AD) test indicates that only the Average Change Method ($p = 0.275 > 0.05$) does not reject the normality assumption ($p < 0.05$) while both Naïve with Drift and Holt's Method reject the null hypothesis at the 5% level of significance for normality. As for the Box-Pierce test, Naïve with Drift ($p = 0.9321$) and Holt's Method ($p = 0.9087$) have better values than the Average Change Method ($p = 0.0002$), leading to greater independence of residuals. Durbin-Watson test values close to 2 indicate less autocorrelation. All models fall in the range between 0 and 2, with Holt's Method (0.35) showing slightly greater degree of autocorrelation than Naïve with Drift (0.42), whereas Average Change Method (0.000027) shows high degree of autocorrelation. With the Breusch-Pagan test (homoskedasticity), Naïve with Drift (p =

0.1168) and Holt's Method (p = 0.0983) have a greater p-values than Average Change Method (p = 0.0093), indicating a better assumption of constant variance. While Average Change Method meets the normality assumption, it does not meet the other assumptions. Naïve with Drift is consistent but neither over nor under-performs. Holt's Linear Trend Method, despite only slightly violating the normality assumption, is the best overall, the only model to pass the independence test, had moderate and non-zero autocorrelation in residuals, and satisfied homoskedasticity. Based on this information, the best model in Part A is Holt's Linear Trend Method.

As per Part B's Anderson-Darling (AD) test, none of the models satisfy the normality assumption entirely (p > 0.05). At the 5% significance level, the null hypothesis was rejected by the ARMA (p = 0.0416), Auto ARIMA (p = 0.0025), ARIMA(1,1,1) (p = 0.0078), and AR (p = 0.0192) model, which are the ones that deviate against normalcy the most. All models demonstrate low to acceptable p-values for residual independence using the Box-Pierce test. In terms of best to worst, ARMA has been followed by AR. Both Auto ARIMA and ARIMA(1,1,1) have marginally lower p-values of 0.3844 and 0.5382. The residual independence test suggests satisfactory independent residuals for all models regardless of difference in values. The Durbin-Watson test suggests that all models exhibit minimal autocorrelation and are consequently within the optimal level (>2), or higher. In this regard ARMA (0.7242) and ARIMA(1,1,1) (0.8216), both indicate very low autocorrelation. While remaining reasonable, AR(0.3709) and Auto ARIMA (0.5284) suggest slightly more residual reliance. The results of the Breusch-Pagan test, ARMA (p=0.1885) performed the best showing that the residuals are homoskedastic (random variation). After ARMA, the assumption was have the same result with AR (p= 0.1642) , and performed ARIMA(1,1,1) (p=0.1216), then Auto ARIMA (p=0.0996) which get an appropriate result however slightly less strong compared to the others models.

It's important to widen even if the ARMA (1,0,1) model has the highest statistical validation of the p-value assumption test results including residual independence, homoskedasticity, and both error metrics indicating low autocorrelation. The Auto ARIMA model is the superior model in this project, even if the ARMA model is effective in validating theoretical assumptions.

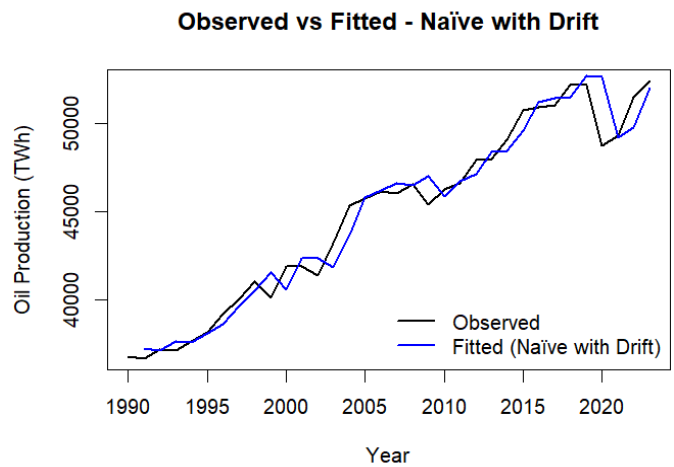
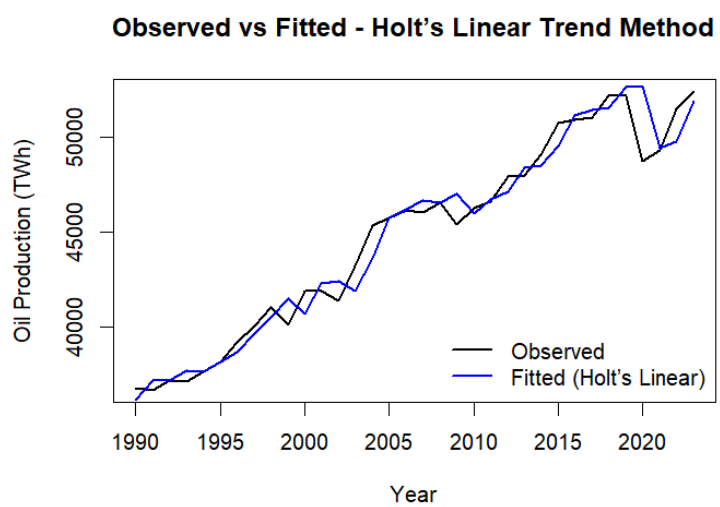
Additionally, all forecasts show bias and represent residual error based on its R-square value. The Auto ARIMA(0,1,0) model was consistently able to show lower

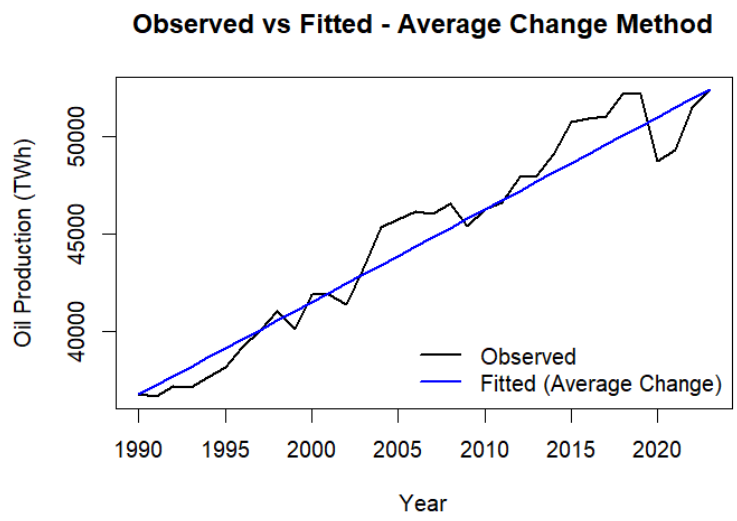
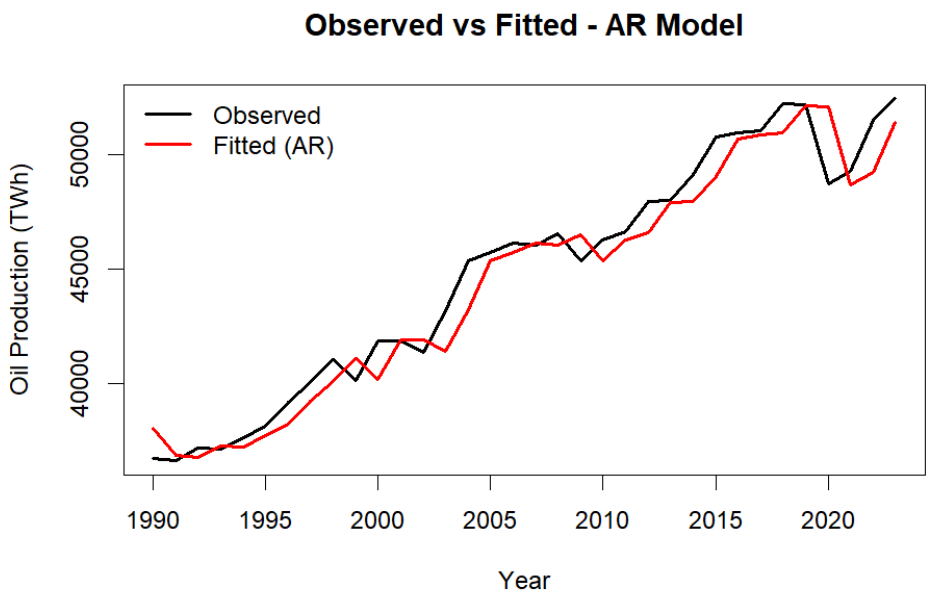
error values in each model category (RMSE, MAE, MAPE, and MASE), which indicates it had more accurate forecasting and less bias overall. Plus, it allowed each forecasting method to choose the best parameters depending on the data, while achieving a fair combination of residual behavior, flexibility, and computational simplicity.

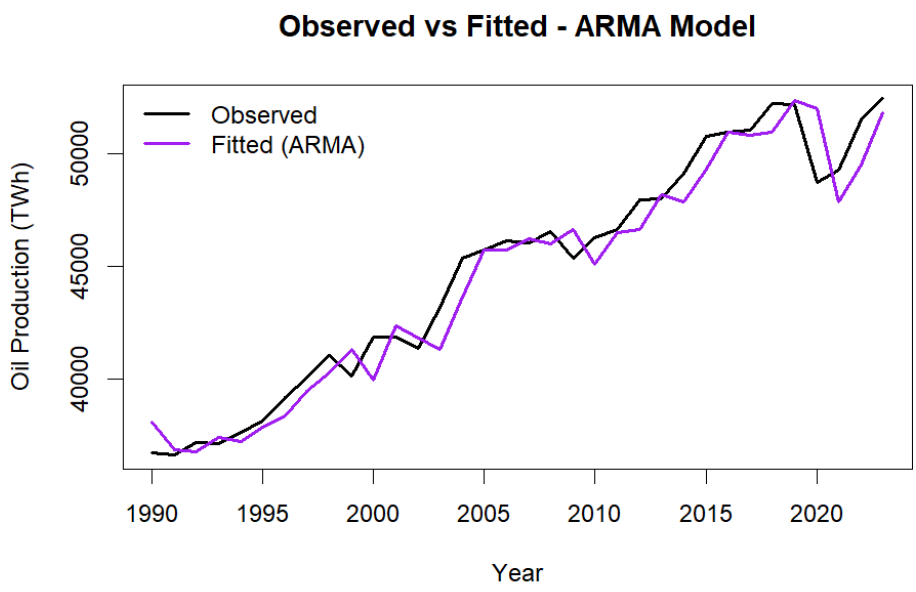
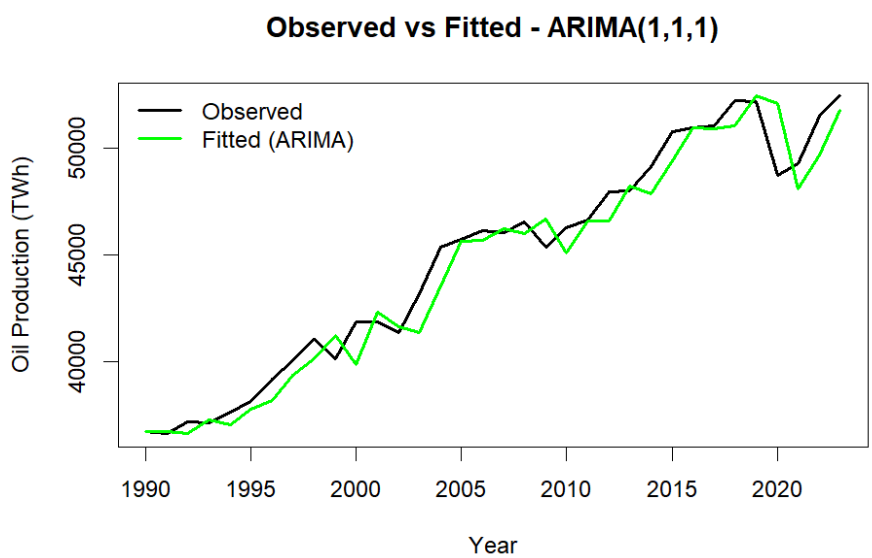
As a result, Auto ARIMA (0,1,0) is the best and most accurate model overall, particularly for practical forecasting regarding global oil production trends. ARMA could end up being the best model based on the assumption tests performed.

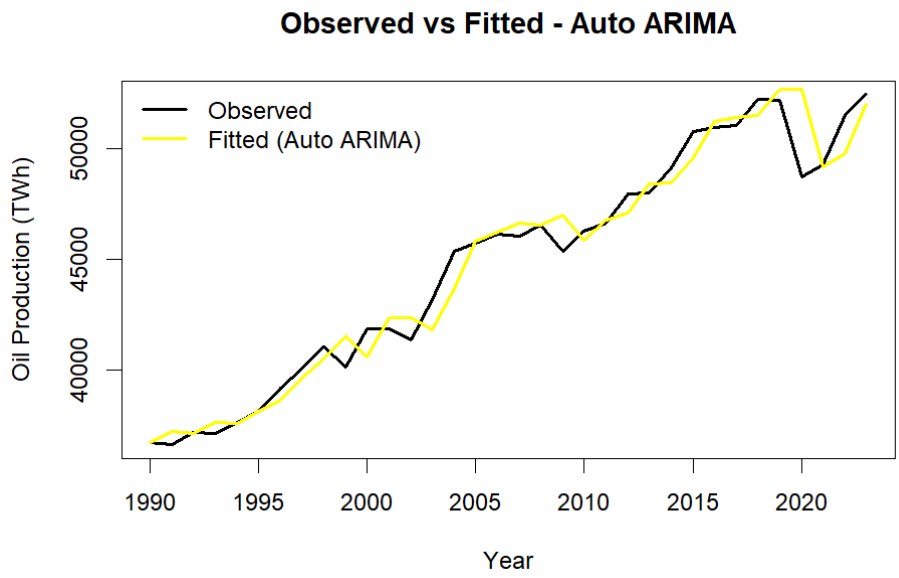
4.2 Analysis On Fitted Plots

Table 4.2 Fitted plot for each univariate time series model.

Model	Fitted Plot
PART A (ETS)	
Naïve with Drift	 **Figure 4.1:** Observed vs fitted values for global oil production (1990–2023) using Naïve with Drift method
Holt's Linear Trend Method	 **Figure 4.2:** Observed vs fitted values for global oil production (1990–2023) using Holt's Linear Trend method

Average Change Method	 **Figure 4.3:** Observed vs fitted values for global oil production (1990–2023) using Average Change method
PART B	
AR (1)	 **Figure 4.4:** Observed vs fitted values for global oil production (1990–2023) using Autoregressive Model

ARMA (1,0,1)	 **Figure 4.5:** Observed vs fitted values for global oil production (1990–2023) using Autoregressive Moving Average Model
ARIMA(1, 1,1)	 **Figure 4.6:** Observed vs fitted values for global oil production (1990–2023) using Autoregressive Integrated Moving Average Model

Auto ARIMA (0,1,0)	 **Figure 4.7:** Observed vs fitted values for global oil production (1990–2023) using Automatic Autoregressive Integrated Moving Average.

The observed vs fitted value plots provide additional insights into how well each forecasting approach captured the trend and variation anticipated in global oil production data from 1990 to 2023.

For Part A, Naïve with Drift and Holt's Linear Trend Method provide closely matched fitting performance as seen in Figure 4.1 and 4.2. Both models do well at representing the general upward trend in the data and are able to capture moderate variations. However, the Naïve with Drift model is more sensitive to recent changes in that it just uses the last observed value and a constant drift. It is simple and responsive, but for when volatility increases, it can diverge significantly with the forecast. In contrast, Holt's method does two volumes of smoothing which are level and trend resulting in a more level fitted line so that Holt's method preserves consistency over time. In the plot, Holt's method provides slightly improved performance over Naïve with Drift during volatile periods by absorbing some of the noise.

Average Change Method is a mathematically simple model, but offers a more linear and less adaptive fit. The fitted line consistently rises, it will consistently be lower

than or higher than actual production when the data fluctuates. It will do this because it uses a consistent average rate of change across the years, giving it more rigidity. In Figure 4.3, it cannot capture sudden changes particularly after 2010, therefore it is more inflexible than the other methods when the series displays structural breaks and non-linear trends.

Overall, the Holt's Linear Trend Method seems to be a good compromise of flexibility and smoothness, with the ability to trend upward or lower, while still capturing more gradual trends and fluctuations. As such, it is the most reliable model for fitting this time series based simply on graphical analysis. Naïve with Drift provides a close competing model but allows potential to overfit or underfit relatively recent values in volatile circumstances. Average Change could be used for a rough approximation. However, it is not a succinct and more appropriate method when dealing with dynamic data such as global oil production.

However, The observed vs fitted plots for Part B give a visual representation of how well each univariate time series model captures the past trend and variation in global oil production from 1990 to 2023. All four models demonstrate different behaviors in respect to flexibility, response and the accuracy of fit, even if all generally follow a similar upward trend.

In cases when there are sharp increases or sharp decreases, the fitted line from the AR(1) model in Figure 4.4 tends to lag behind the overall data movement. It is a simple way of incorporating historical data and is a straightforward approach to take but less flexible when the series experiences sudden movements. It is reasonably good but is somewhat inflexible and may not perform well in these more dynamic periods in the dataset. In contrast, the ARMA(1,0,1) model appears, on the whole, to be more consistent with the actual observations in Figure 4.5. The addition of a moving average component allows the overall model to be more flexible/ responsive, as one adjusts for previous errors. Specifically, the fitted line appears to resemble the actual data much more closely during transitional years, suggesting that the ARMA model does a better job of capturing short-run movements than the AR model.

When using differencing to handle non-stationary data, the ARIMA(1,1,1) model shown in Figure 4.6 substantially improves the overall fit. The graphic illustrates how

well ARIMA captures the mid-range fluctuations and the long-term trend, and only a relatively slight error. ARIMA appears to be a reasonable option when the data show both trend and volatility, as it provides a nice compromise between smoothness and flexibility. Despite the less complex structure, the Auto ARIMA(0,1,0) model (Figure 4.7), produces a fitted line that closely follows the actual values over the duration of the series. It is able to respond to changing trends in oil production because it has the drift component and with its ability to choose the parameters automatically. Among all the models in Part B, Auto ARIMA has the best representative fit and the best fit of the trends and variations in the data.

Overall, Auto ARIMA appears to provide the most accurate and consistent fit from a visual perspective amongst the four models in Part B. This is then followed by ARMA and ARIMA(1,1,1), which provided good variance characteristics and preserved the trend of the data. The AR model was the least accurate even though it was the simplest and may not have been sufficient to develop complex patterns of oil production. This graphical review reinforces the conclusion that Auto ARIMA is the preferred model in this project for forecasting.

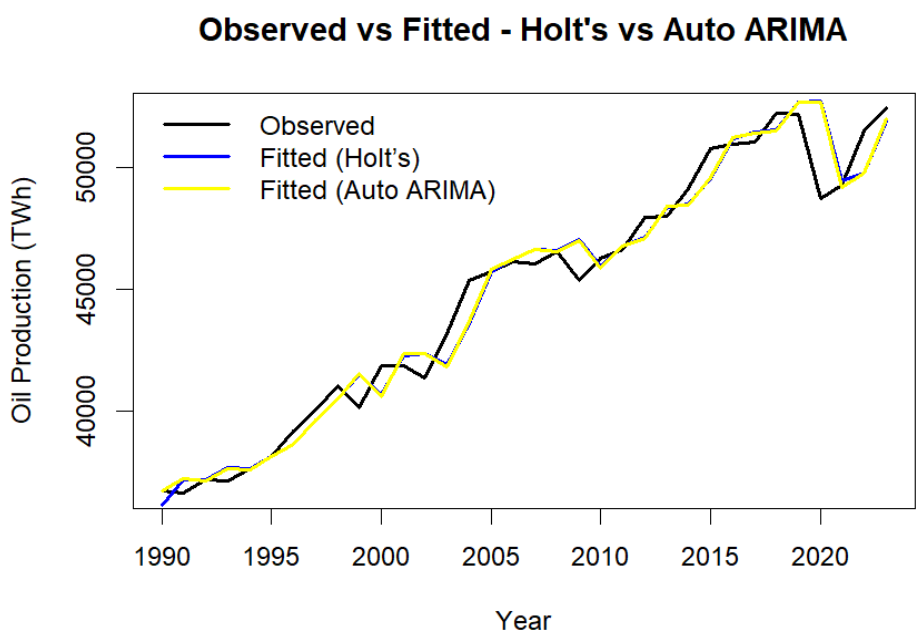


Figure 4.8: The plot of observed and the fitted time series of two best model (Auto ARIMA vs Holt's)

The fitted values of the two best-fitting models, Auto ARIMA and Holt's Linear Trend, are compared to the actual global oil production data from 1990 onwards in the figure 4.8 above. Holt's model takes smooth long-term changes, or level and trend smoothing, into account, and represents this process better than the remaining models, since both approaches are able to replicate the general growth trend. However, due to its linear structure, it struggled during the volatile past. Auto ARIMA (0,1,0), on the other hand, captures more accurately the temporal dependencies and is well tuned into fluctuations. Ultimately Auto ARIMA is better able to predict trends in oil production because its fitted values were closer to the actual data.

4.3 Analysis On Error Measure

Table 4.3 Error measures

MODEL	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	
PART A (ETS)								
Naïve with Drift	-2.204834e-12	1054.102	728.968	-0.02163159	1.601971	0.8459948	-0.01417479	
Holt's Linear Trend Method	3.191023	1042.979	714.344	-0.00561818 7	1.576763	0.8290232	0.01883826	
Average Change Method	247.2754	1205.574	959.5108	0.4146	2.088	1.1135	-	
PART B								
AR (1)	424.11	1154.72	882.52	0.90	1.95	1.02	0.0145	
ARMA (1,0,1)	340.15	1133.74	871.08	0.71	1.93	1.01	-0.1366	
ARIMA(1,1,1)	401.48	1115.10	839.98	0.89	1.85	0.97	-0.1584	
Auto ARIMA (0,1,0)	1.07	1038.50	708.59	-0.02	1.56	0.82	-0.0147	

Table 4.3 shows the measure of error produced from each of the three univariate models in Part A which is the Naïve with Drift, Holt's Linear Trend Model, and the Average Change method. The measure of errors includes the Mean Error (ME), root mean square error (RMSE), Mean Absolute Error (MAE), Mean Percentage Error (MPE), Mean Absolute Percentage Error (MAPE), Mean Absolute Scaled Error (MASE), and first order autocorrelation (ACF1) of residuals, and these error metrics were used to assess the forecasting performance of the model. In all of the metrics evaluated, the Naïve with Drift and Holt's Linear Trend Method models perform better than the Average Change Method. The RMSE, MAE, MAPE, and MASE for the Naïve and Holt models are all appreciably lower than the Average Change suggesting greater accuracy of next-period predictions and lower magnitude of errors. Among the error values between Naïve methods and Holt's methods are fairly closely matching including RMSE (1054.10 vs 1042.98), MAE (728.97 vs 714.34), MAPE (1.60 vs 1.58), and MASE (0.8460 vs 0.8290). ME and MPE are slightly better in terms of showing less bias and better centering in Holt's Linear Trend Method. Holt's model also demonstrates a positive ACF1 (+.0188) suggesting slightly more autocorrelation than the Naïve model (-0.0142), but both values indicate overall low autocorrelation. In contrast, the Average Change Method describes the worst performance with much higher error values across the board including RMSE (1205.57), MAE (959.51), MAPE (2.088), and MASE (1.1135). This indicates that the Average Change Method is not as well suited to capturing the dynamic trends in global oil production. To summarize, the Holt's Linear Trend Model is easily the best model for Part A on smaller size errors, low bias, and also the little sensitivity of the model when forecasting in all metrics. The Naïve model would be a competitive model but Holt's is better suited for oil production time series data as a forecasting model and demonstrates better accuracy and robustness.

Similarly, the error measures for the four univariate time series models AR(1), ARMA(1,0,1), ARIMA(1,1,1), and Auto ARIMA(0,1,0) are summarized in Table 4.3. Among these, Auto ARIMA demonstrates the best forecasting performance across all error metrics. It achieves the lowest values for RMSE (1038.50), MAE (708.59), MAPE (1.56), and MASE (0.82), highlighting its superior accuracy in predicting global oil production. Furthermore, Auto ARIMA also records the lowest ME and MPE, suggesting minimal forecasting bias and better centering of its predictions around the actual observed values.

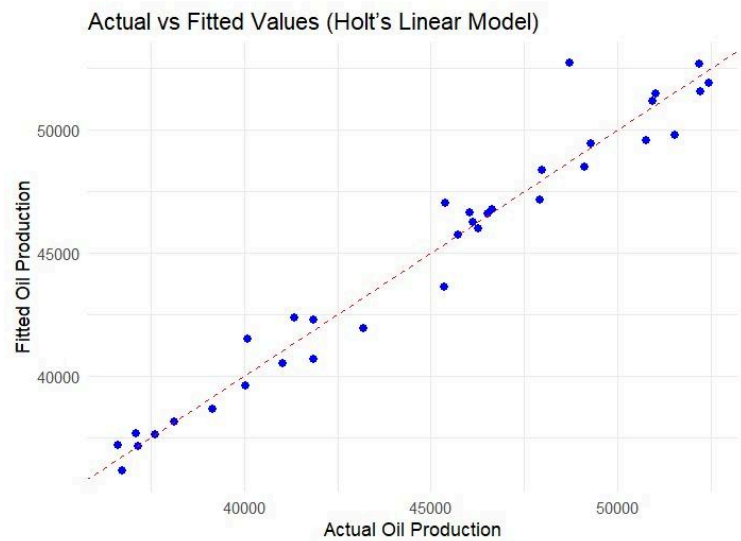
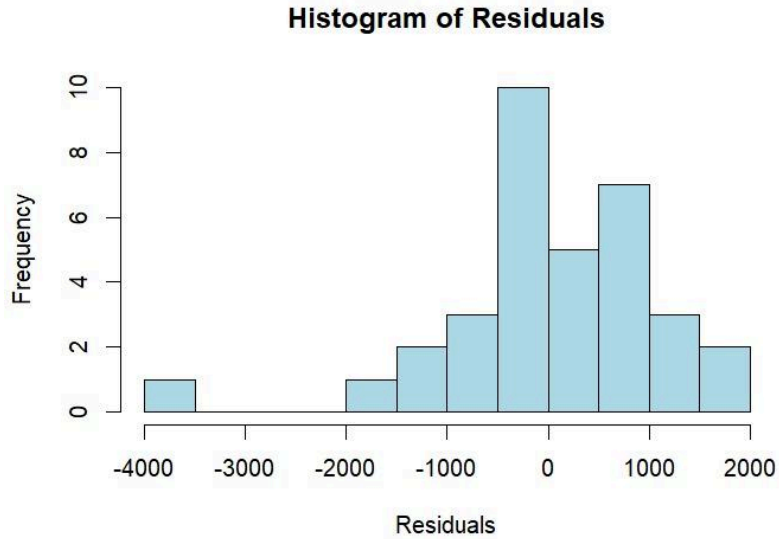
In terms of comparative performance, the ARIMA(1,1,1) and ARMA(1,0,1) models follow closely behind Auto ARIMA. Both produce relatively competitive error values and appear to be reasonably accurate, but still fall short in precision when compared directly. Their residual autocorrelation values ACF1 of –0.1584 for ARIMA(1,1,1) and –0.1366 for ARMA indicate some lingering dependency in residuals, which can slightly reduce reliability in long-term forecasting.

The AR(1) model, however, performs the least favorably among the four. It produces the highest RMSE (1154.72) and MASE (1.02), suggesting a higher magnitude of prediction error and less suitability for modeling the dynamics of oil production data. This underperformance may be attributed to the model's simplicity, as it lacks both moving average terms and differencing to handle underlying trends.

In summary, Auto ARIMA (0,1,0) stands out as the most robust and accurate model in Part B. Its ability to automatically optimize model parameters, account for trends through differencing, and maintain low error magnitudes makes it the most suitable forecasting tool for this time series dataset. It balances simplicity and predictive power effectively, outperforming the other candidate models in both accuracy and consistency.

4.4 Analysis On Residuals Plot

Table 4.3 Qualitative residuals check of the residuals from the best model

Model	Residuals Plot
PART A	
Holt's Linear Trend Method	 **Figure 4.9** Actual vs Fitted Values  **Figure 4.10** Histogram of Residuals

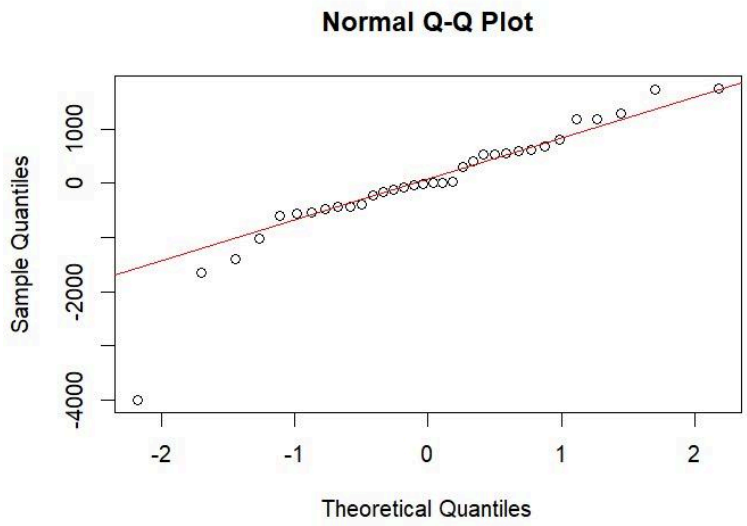


Figure 4.11 Q-Q Plot I

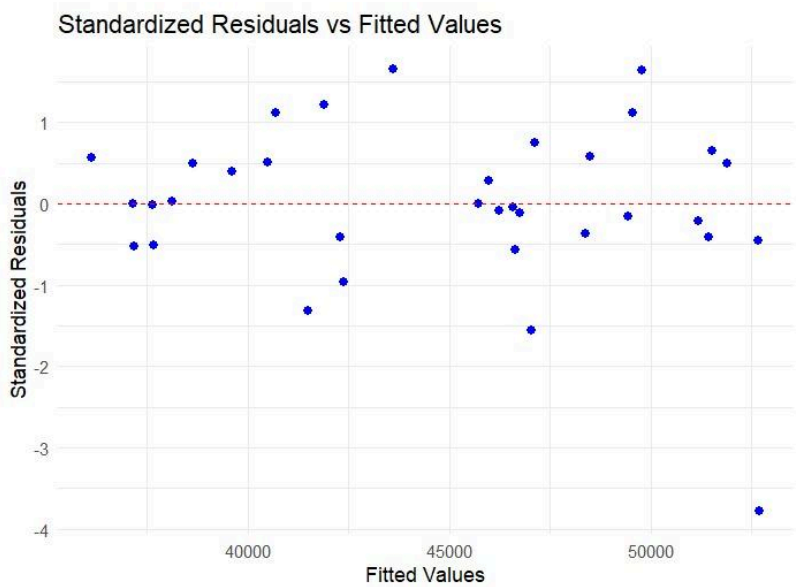


Figure 4.12 Standardized Residuals vs Fitted

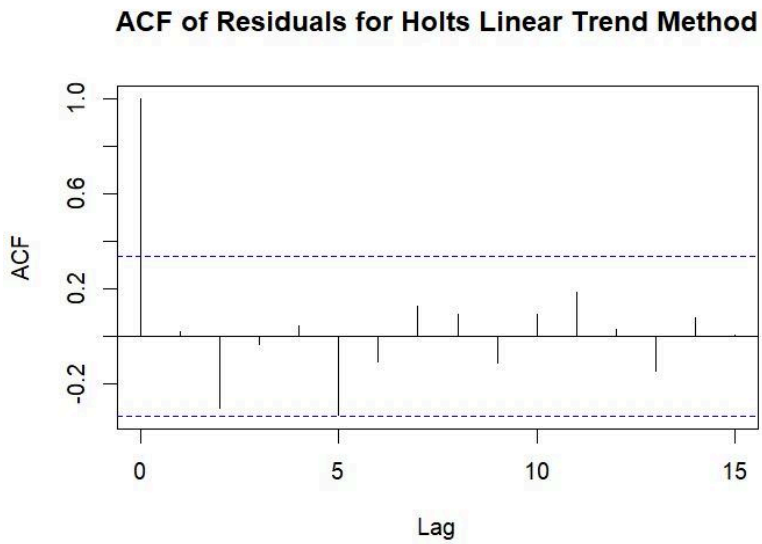


Figure 4.13 ACF of Residuals

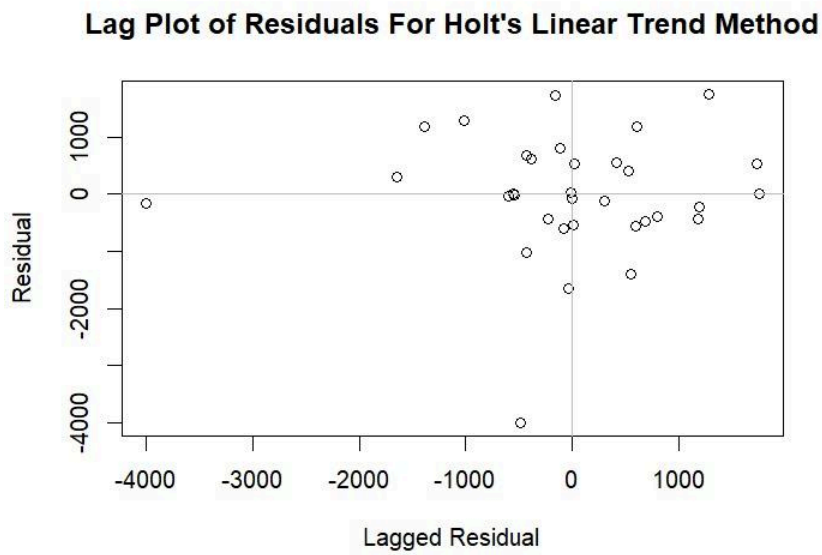


Figure 4.14 Lag Plot of Residuals

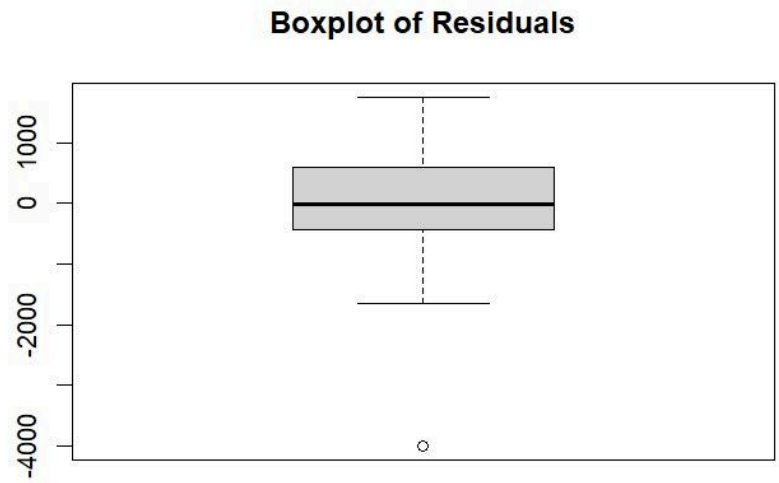


Figure 4.15 Boxplot of Residuals

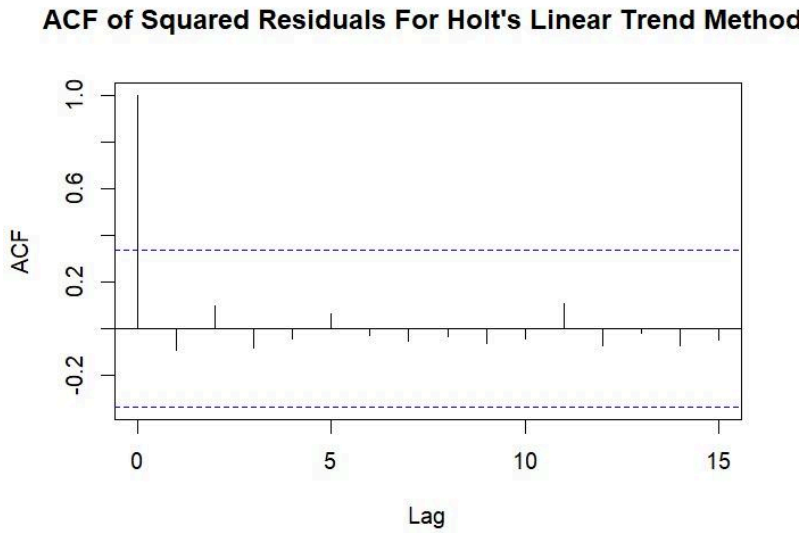


Figure 4.16 ACF of Squared Residuals

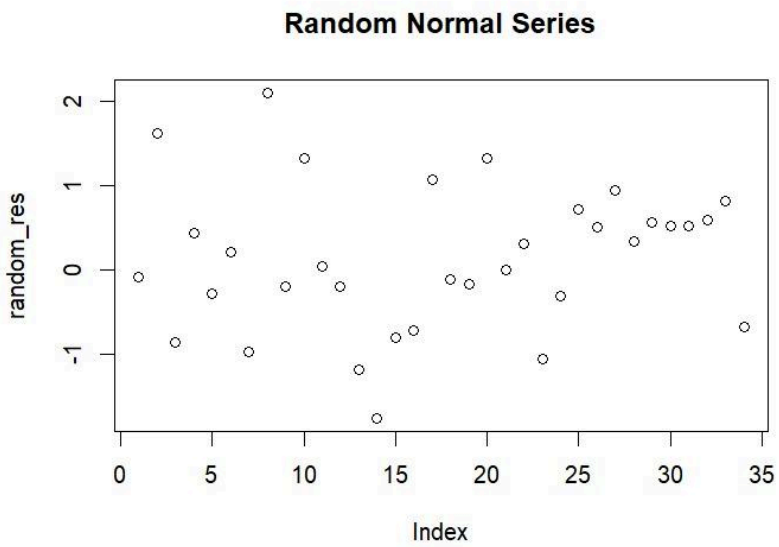


Figure 4.17 Plot of Random Normal Residual Series

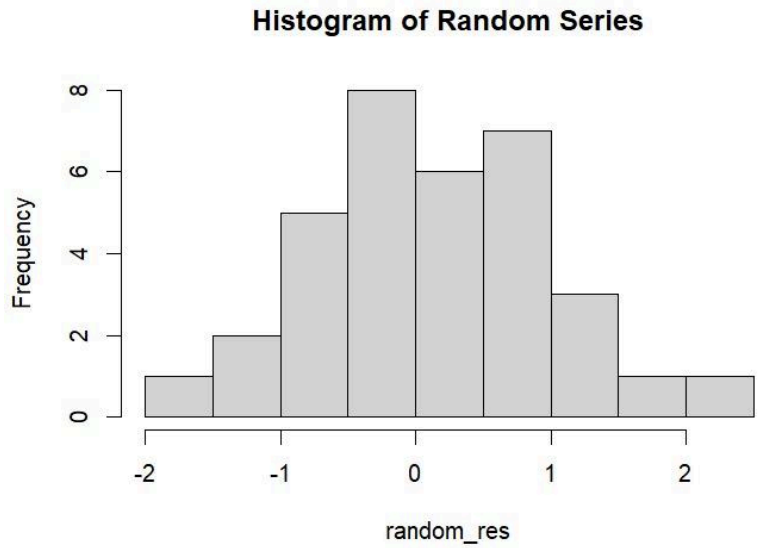


Figure 4.18 Histogram of Random Normal Residuals

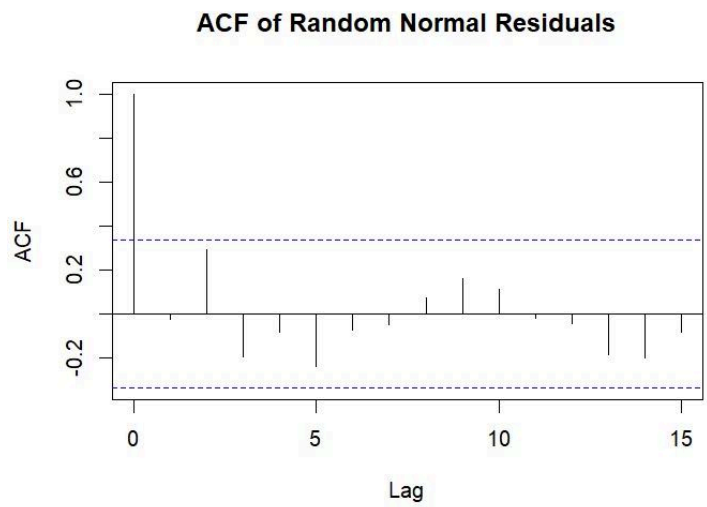


Figure 4.19 CF of Random Normal Residuals

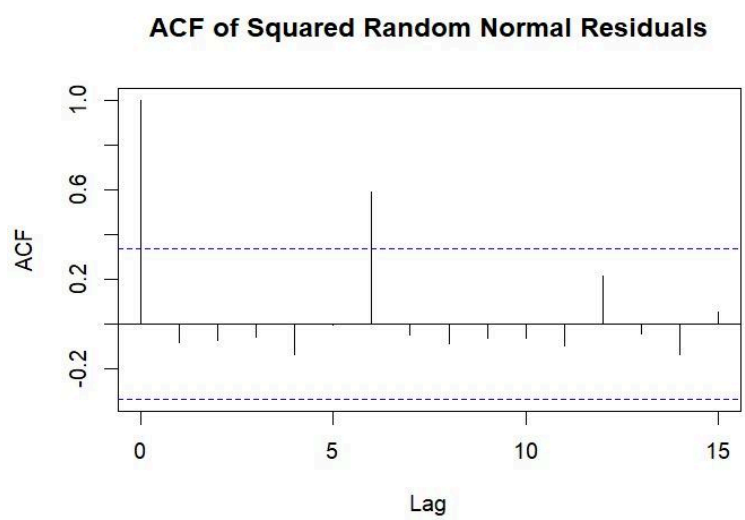


Figure 4.20 ACF of Squared Random Normal Residuals

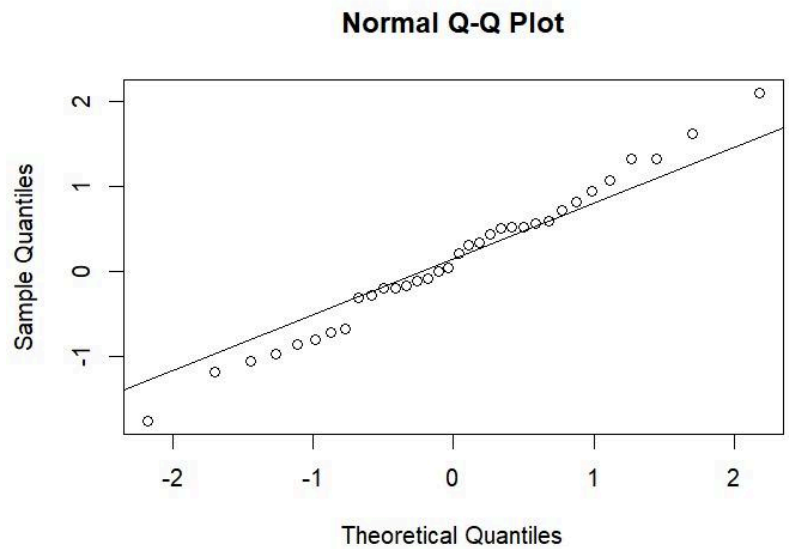


Figure 4.21 Q-Q Plot of Random Normal Residuals

PART B

Auto ARIMA (0,1,0)

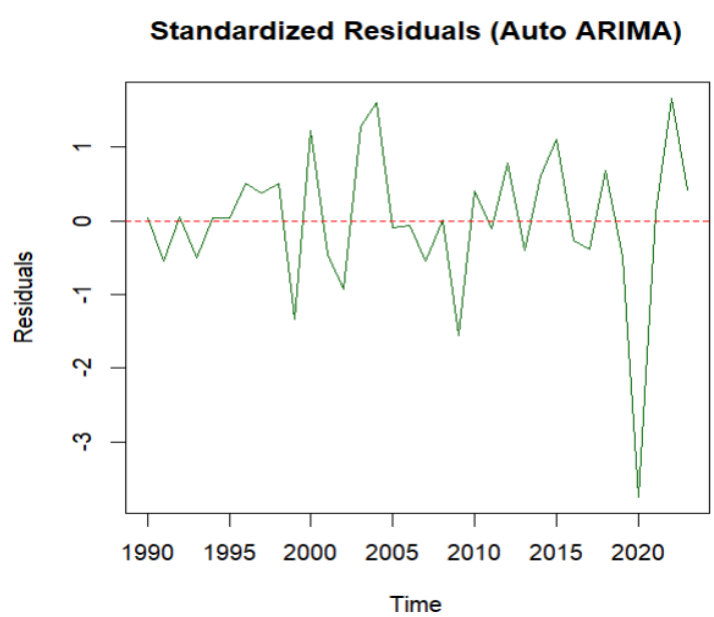


Figure 4.22 Standardized Residual

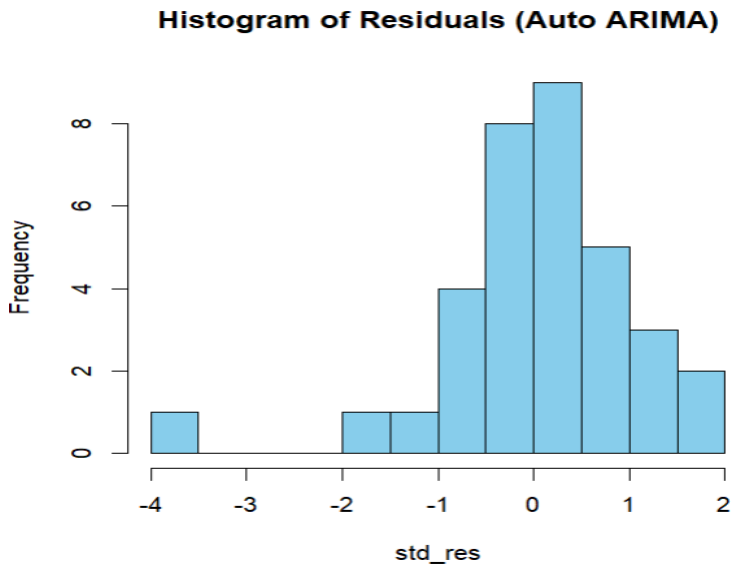


Figure 4.23 Histogram of Residuals

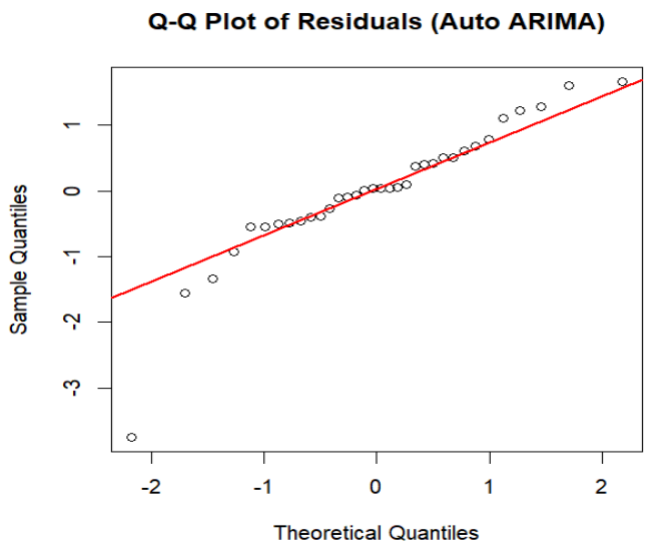


Figure 4.24 Q-Q Plot of Residuals

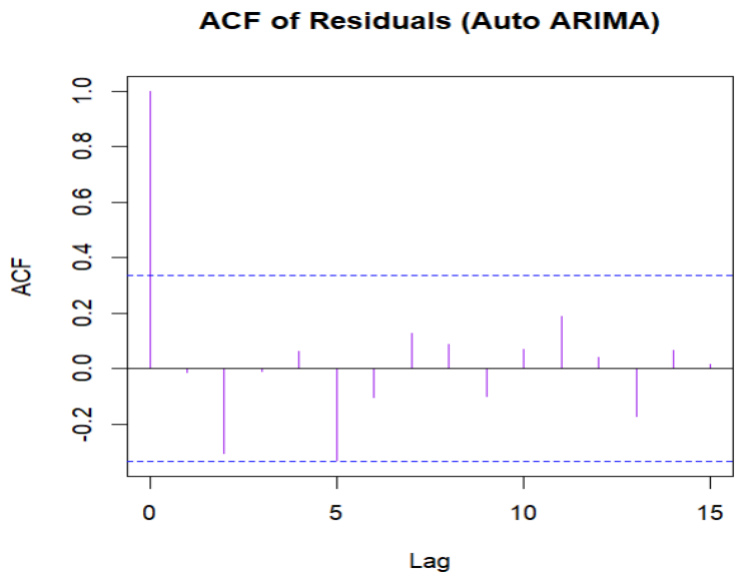


Figure 4.25 ACF plot of residuals

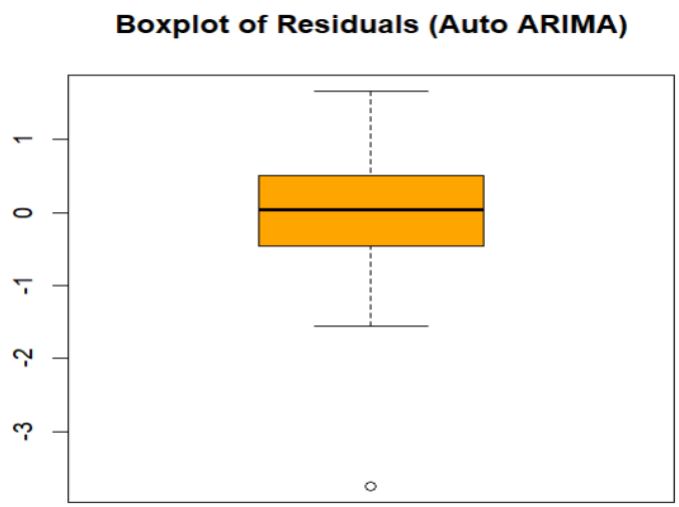


Figure 4.26 Boxplot of Residuals

Residual analysis was performed to evaluate the fit of the Holt's Linear Trend model for forecasting global oil commodity production. It included graphical diagnostics as well as tests for statistical assumptions such as normality, independence, constant variance, and for autocorrelation.

The Q-Q plot (Figure 4.11) compares the quantiles of the residuals to a theoretical normal distribution. In a good fitting model, the residuals should lie close to a diagonal line. In the plot most points lie close to the diagonal line, indicating the residuals are approximately normally distributed. This is also supported by the results from the Anderson-Darling and Kolmogorov–Smirnov tests which suggested that there were no strong violations to the assumption of normality.

The histogram of the residuals (Figure 4.10) also supports this conclusion. The residuals have a bell shape centered on the zero as there is no evident skewness or multimodal distribution. This is important because it demonstrates that the error terms are symmetrically distributed which further supports the assumption that the residuals are distributed along a roughly normal distribution pattern.

Next, the plot of the standardized residuals against the fitted values (Figure 4.12) is checked to test for homoscedasticity as the points are randomly dispersed around a horizontal at zero which also does not indicate a funnel effect or some other curvature pattern. This reinforces the conclusion of constant variance about the residuals across the values of the fitted values. The Breusch-Pagan test is used for heteroscedasticity and presents significant results as the test does not reject the null hypothesis.

The Durbin-Watson test, together with the ACF plot of the residuals (Figure 4.13), were performed to check for the absence of autocorrelation. The ACF plot indicates that there are no significant autocorrelations after lag 1. The Durbin-Watson statistic was also close to 2 for all residuals. Both the ACF plot and the Durbin-Watson statistic suggest that the residuals appear to be serially uncorrelated. The Ljung-Box test of the residuals also contributes to the notion that the residuals can be treated as white noise. The lag plot of residuals (Figure 4.14) also shows no visible trend or structure between the residuals and the lagged values, which supports the independence assumption. The boxplot of the residuals (Figure 4.15) identifies one mild outlier, and most of the residuals

lie within the interquartile range, which suggests that the model is robust against influential observations. The overall shape is symmetric and centered around zero.

Lastly, the ACF of squared residuals (Figure 4.16) suggests that the residuals are not conditionally heteroscedastic. There is no obvious autocorrelation in the squared residuals, thus the residuals of the model are stationary and do not appear to exhibit volatility clustering.

For benchmarking purposes, a set of random normal residual data (Figures 4.13 to 4.21) was created and analysed at the same time. Plots such as these provide a reference baseline confirming that the Holt model residuals are very similar to what would be expected of a simple white noise process.

PART B

The standardized residuals of the Auto ARIMA model, displayed in Figure 4.22, randomly fluctuate around zero, with one outlier noted towards the end of the year. This randomness with no apparent trend indicates that the residuals from this model are independent.

The histogram of the residuals is shown in Figure 4.23, and is approximately symmetric around zero. The near symmetry suggests no meaningful skewness or kurtosis, and that the model residuals are approximately normally distributed. The Q-Q plot shown in Figure 4.24 provides further evidence, with a satisfactory fit of the residual points to a theoretical normal distribution closely matching the diagonal reference line.

The ACF plot of residuals in Figure 4.25 shows no evidence of autocorrelation, other than the bounds of tolerance, indicating that there is no serial correlation among the residuals and that the model has accounted for the time dependencies in the sequence. Finally, the boxplot of residuals in Figure 4.26 supports the results of normality and constant variance. The median is almost zero and even though there is one outlier, the spread is fairly evenly spaced.

In conclusion, all of the diagnostic tests, along with each other, provide evidence that Auto ARIMA (0,1,0) is a good fit to the data and that it met the necessary assumptions of a reliable time series model.

5. Prediction or Forecasting

Based on the recent analysis, Model 2: Holt's Linear Trend Method and Model 4: Auto ARIMA (0,1,0) are the best models for forecasting oil production. The 12-step ahead forecasts starts from the year 2024 to assess any possible future trends. The forecasted values from year 2024 to year 2035 are listed in Table 5.1 and Table 5.2:

Table 5.1 Forecasted Oil Production (TWh) For Holt's Linear Method Model with 80% and 95% Confidence Intervals from 2024–2035

Year	2024	2025	2026	2027	2028	2029
Forecast (TWh)	52889.78	53379.75	53869.73	54359.71	54849.69	55339.66
Lower 80%	51466.82	51427.99	51504.49	51643.14	51822.22	52030.31
Upper 80%	54312.73	55331.52	56234.97	57076.29	57877.17	58649.03
Lower 95%	50713.56	50394.79	50252.41	50205.07	50219.57	50278.45
Upper 95%	55065.99	56364.72	57487.06	58514.35	59479.81	60400.89

Year	2030	2031	2032	2033	2034	2035
Forecast (TWh)	55829.64	56319.62	56809.60	57299.57	57789.55	58279.53
Lower 80%	52260.55	52508.39	52770.67	53045.07	53329.83	53623.58
Upper 80%	59398.75	60130.87	60848.55	61554.10	62249.30	62935.50
Lower 95%	50371.18	50490.84	50632.58	50792.86	50968.99	51158.86
Upper 95%	61288.12	62148.42	62986.63	63806.31	64610.14	65400.22

Table 5.2 Forecasted Oil Production (TWh) for Auto ARIMA (0,1,0) with 80% and 95% Confidence Intervals from 2024–2035

Year	2024	2025	2026	2027	2028	2029
Forecast (TWh)	52908.15	53384.07	53860.00	54335.92	54811.84	55287.76
Lower 80%	51536.29	51443.98	51483.87	51592.21	51744.28	51927.42
Upper 80%	54280.01	55324.17	56236.12	57079.63	57879.40	58648.11
Lower 95%	50810.08	50416.95	50226.03	50139.77	50120.41	50148.56
Upper 95%	55006.22	56351.19	57493.96	58532.06	59503.27	60426.97

Year	2030	2031	2032	2033	2034	2035
Forecast (TWh)	55763.69	56239.61	56715.53	57191.45	57667.38	58143.30
Lower 80%	52134.10	52359.42	52599.97	52853.27	53117.45	53391.05
Upper 80%	59393.27	60119.80	60831.10	61529.64	62217.31	62895.55
Lower 95%	50212.71	50305.37	50421.32	50556.77	50708.86	50875.37
Upper 95%	61314.66	62173.85	63009.75	63826.14	64625.89	65411.23

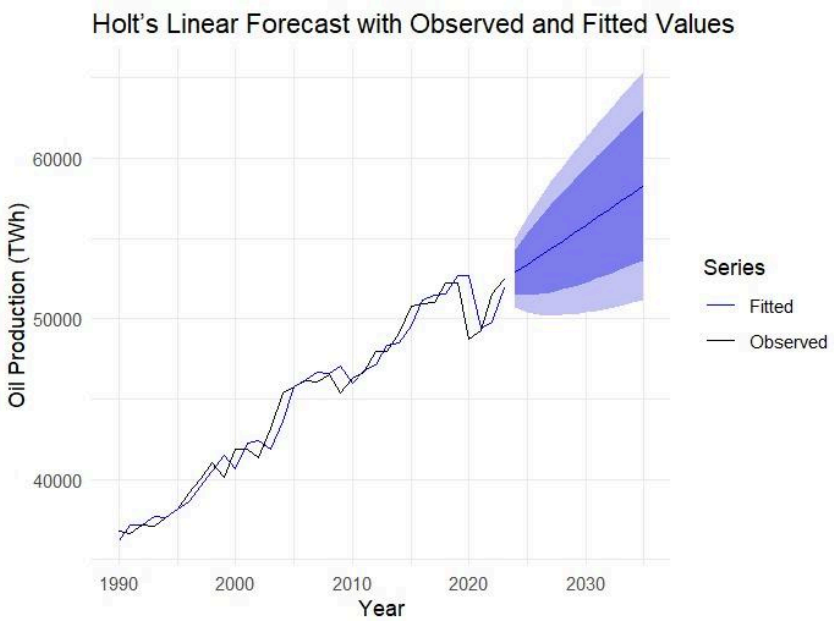


Figure 5.1 Holt's Linear Model Forecast with Observed and Fitted Oil Production Values

6. Conclusion

This study is about a univariate time series model and prediction on global oil production on past data from 1990-2023. Two forecasting frameworks have been used such as Exponential Smoothing (ETS) approaches, which were analyzed in Part A, whereas ARIMA-based Box-Jenkins models were analyzed in Part B. It was intended that their performance would be compared and assumptions of the models would be evaluated. The model that would give accurate outcomes during future forecasting to the year 2035 would be identified.

Out of the ETS methods, the Linear Trend Method by Holt has been found to be the best one since it provides smooth and interpolable forecasts. It also had better scores in terms of bias, residual behavior, and error statistics such as the RMSE and MAPE. But it is its assumption of a linear trend that makes it not very flexible when there is a time of structural change.

Contrastingly, the Auto ARIMA(0,1,0) model was the finest performing model overall among the ARIMA models. It showed the smallest forecasting errors (RMSE = 1038.50, MAE = 708.59, MAPE = 1.56), did not fail the majority of residual diagnostic tests, and revealed underlying dynamics of oil production accurately. Its capability to automatically choose parameters depending on the properties of the datasets has enabled it to allow superior flexibility, especially in predicting timeframes of moderate or unpredictable growth.

The forecast trend between the two models was very similar because the global oil production has kept increasing steadily all throughout the study period. But Auto ARIMA had narrow confidence intervals and good residual characteristics thereby being more robust and statistically strong.a

In conclusion, though Holt Linear Trend Method and Auto ARIMA are valid for the long-term oil production forecasting, Auto ARIMA can be said as the best model to use because it performed better based on all the criteria considered in its evaluation. This model is capable of giving policymakers, energy analysts and any other interested stakeholder a viable framework through which to predict the global oil supply in the future particularly in situations where external predictors are poor or unavailable.

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